DOES HYPERBOLIC GEOMETRY HELP GRAPH-BASED SENTENCE POOLING? A CONTROLLED STAGE-WISE STUDY

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ABSTRACT

Hyperbolic space embeds hierarchies with far less distortion than Euclidean space, and token states are often argued to be approximately hierarchical. That reasoning has motivated hyperbolic variants of many architectures. We test it carefully on one: GLOT (Mantri et al., 2026), a graph-based sentence pooler that builds a latent token graph over a frozen backbone, refines it with a GNN, and aggregates with a learned readout.

GLOT has exactly three places where geometry enters, so we build a hyperbolic counterpart of each and measure them *separately* rather than as a bundle: a Poincaré geodesic token graph (**A**), an Einstein gyro-midpoint readout (**B**), and Möbius message passing (**C**). All eight combinations run under an equal search budget over the original model's own hyper-parameter grid, with 15 paired seeds, on three GLUE tasks, a decoder backbone, and a noise-robustness diagnostic.

The answer is *partly, and only in one component*. Hyperbolic message passing (C) is the one positive: $+1.42$ MCC on CoLA with $13/15$ seeds, significant on a paired t-test and an exact sign test, and A+C is the best arm on STS-B. The gain does not generalise — C is slightly negative on MRPC, and on CoLA it is not separable from a control that deletes the graph entirely. The gyro-midpoint readout (B) is *reliably harmful*: 0–1 of 15 seeds positive on STS-B, surviving Bonferroni, and reproducing on a decoder backbone where it costs 3.6 Spearman. The hyperbolic graph (A) is flat everywhere. Because B and C carry opposite signs and similar magnitudes, a study reporting "hyperbolic pooling" as a single switch would have averaged them to approximately nothing.

We also question the premise. The usual justification — low measured δ-hyperbolicity of token activations — is an artifact of norm outliers: on the outlier-invariant angular metric ModernBERT's δ is 0.2127 against BERT's 0.2114, statistically indistinguishable, while the Euclidean estimate differs by two orders of magnitude. Measured tree-likeness is a poor guide to whether hyperbolic layers will help, which is consistent with what we observe.

1 INTRODUCTION

Reducing a language model's token states to a single vector is a prerequisite for most sentence-level tasks. Standard poolers — mean, max, `[CLS]` — treat tokens as an unordered set. A recent line of work instead frames pooling as *relational learning followed by aggregation*: build a latent graph over tokens, refine it with a GNN, then aggregate. GLOT (Mantri et al., 2026) is a strong instance, reporting competitive GLUE and MTEB results with $20\times$ fewer trainable parameters than comparable methods while keeping the backbone frozen. Our experiments confirm the premise it rests on: on the authors' own noise diagnostic, GLOT beats every set-based pooler by 14–35 points in every seed we ran.

This paper asks what *else* that architecture can support. Token states form an approximately hierarchical structure, and hyperbolic space embeds hierarchies with far less distortion than Euclidean space, so it is a natural candidate for each of GLOT's three components. We therefore build hyperbolic counterparts of all three — the graph, the readout, and the message passing — and measure them separately rather than as a bundle. That separation turns out to matter: two of the three move in opposite directions, and a study that treated "hyperbolic pooling" as one switch would have reported their average and concluded nothing.

Contributions.

1. **A stage-wise hyperbolic extension** (§6). Under an equal-budget search over GLOT's own grid with 15 paired seeds, hyperbolic message passing (Stage C) gives $+1.42$ MCC on CoLA with $13/15$ seeds positive, significant on both tests, and A+C is the best arm on STS-B; on MRPC it is slightly negative, so the effect is task-dependent. The hyperbolic readout (Stage B) is significantly harmful on both an encoder and a decoder backbone. Reporting "hyperbolic" as a single factor would have cancelled these out.
2. **GLOT is under-reported** (§4). The published numbers fix seed 42 and take the best τ per task on the split they report. At that seed, over that grid, we reach 50.52 MCC on CoLA against a reported 47.49, and 59.57 on RTE against 59.21.
3. **Porting GLOT to a new backbone** (§5). An absolute cosine threshold produces very different graphs on different backbones — density 0.10 on BERT, 0.99 on RoBERTa. Accuracy is nonetheless unchanged, so the method transfers as-is; we show that the two natural "corrections" (density-matching, mean-based norm rescaling) are the interventions that actually cost accuracy, and give the measurements a practitioner needs to avoid them.

2 Three hyperbolic constructions

Let $H = (x_1, \ldots, x_L)$ be the token states of a frozen backbone. GLOT forms

$$A_{ij} = \mathbf{1}[\,\cos(x_i, x_j) > \tau\,], \tag{1}$$

applies K graph-attention layers with jumping-knowledge concatenation, and aggregates the refined states with a learned readout. Only the pooler trains. Geometry enters in exactly three places — how the graph is formed, how messages are combined, and how the refined states are aggregated — so we build a hyperbolic counterpart of each and vary them independently.

Preliminaries. We work in the Poincaré ball $\mathbb{B}^d_c = \{x \in \mathbb{R}^d : c\|x\|^2 < 1\}$ of curvature $-c$, with Möbius addition $\oplus_c$, and the maps

$$\exp_0^c(v) = \tanh\!\big(\sqrt{c}\,\|v\|\big)\,\frac{v}{\sqrt{c}\,\|v\|}, \qquad \log_0^c(x) = \operatorname{artanh}\!\big(\sqrt{c}\,\|x\|\big)\,\frac{x}{\sqrt{c}\,\|x\|}, \tag{2}$$

and geodesic distance $d_c(x, y) = \frac{2}{\sqrt{c}} \operatorname{artanh}\big(\sqrt{c}\,\|{-x} \oplus_c y\|\big)$.

Curvature is searched in a scale-free parameterisation. Curvature and feature scale are not separately identifiable here: d_c depends on $\sqrt{c}\,\|x\|$, so doubling the features and quartering c leaves every distance unchanged. Searching c directly therefore wastes most of the grid. We instead condition features to unit mean norm and search the *effective radius* $r_{\text{eff}} = \sqrt{c}\,\mathbb{E}\|x\|$, which with unit features gives $c = r_{\text{eff}}^2$; we sweep $r_{\text{eff}} \in \{0.5, 1, 1.5, 2, 3\}$, spanning the near-Euclidean regime to strong curvature. This matters: with raw BERT token norms (≈ 14.7), $\exp_0^c$ saturates to radius 1.0 in `float32` and every absolute distance threshold yields an empty graph (Appendix H).

Stage A: Poincaré geodesic graph. Replace Eq. 1 with $A_{ij} = \mathbf{1}[\,d_c(\exp_0^c(\tilde{x}_i), \exp_0^c(\tilde{x}_j)) < \rho_q\,]$, where $\tilde{x}$ is the conditioned feature and ρ_q is the q-quantile of off-diagonal pairwise distances. Using a quantile rather than an absolute radius makes realised density q by construction, so Stage A and the cosine baseline can be compared at *matched sparsity* and the contrast is about geometry rather than about how many edges survive.

Stage B: Einstein gyro-midpoint readout. Replace the readout's weighted Euclidean sum with the Einstein midpoint, computed in Klein coordinates $k = 2x/(1 + c\|x\|^2)$ with Lorentz factors $\gamma_k = (1 - c\|k\|^2)^{-1/2}$:

$$m_K = \frac{\sum_i a_i \gamma_{k_i} k_i}{\sum_i a_i \gamma_{k_i}}, \tag{3}$$

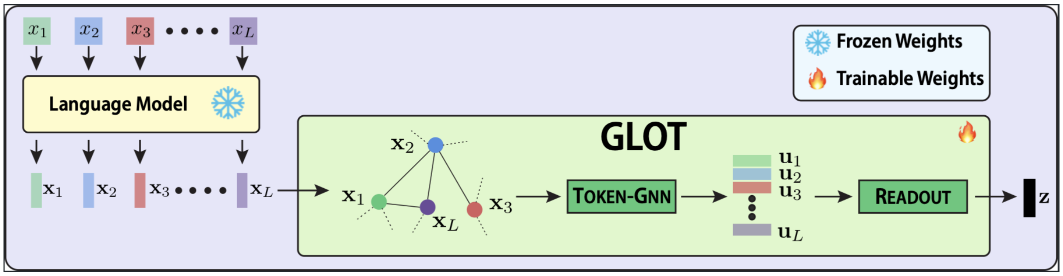

Figure 1: GLOT's pooling pipeline, reproduced from Mantri et al. (2026). Token states $x_1, \ldots, x_L$ from a frozen backbone are thresholded into a latent token graph (Eq. 1), refined by a token-level GNN, and aggregated into a single sentence vector $\mathbf{z}$ by a learned readout; only the shaded block trains. Each of our three modifications replaces exactly one depicted component — Stage A the graph construction, Stage B the readout, Stage C the token-GNN — and we vary them independently rather than as a bundle. The `no_graph` control deletes the edges of the depicted graph while leaving every parameter in place.

for attention weights a_i, mapped back to the ball and then to the tangent space for the task head. Unlike a Euclidean mean this is genuinely curvature-dependent: as $c \to 0$ it recovers the arithmetic mean.

Stage C: Möbius message passing. Replace each GNN layer with a hyperbolic one that lifts to the tangent space to aggregate:

$$h_i' = \exp_0^c\Big(\sigma\Big(\sum_{j\in\mathcal{N}(i)} \alpha_{ij}\, \log_0^c\left(W \otimes_c h_j\right)\Big)\Big), \tag{4}$$

with $W \otimes_c h = \exp_0^c(W \log_0^c(h))$ the Möbius matrix–vector product and α_{ij} either degree-normalised (HGCN) or attention (HGAT).

We run all eight combinations of A, B, C, plus a `no_graph` control that raises τ until essentially no edge survives, leaving parameter count and architecture unchanged and removing only the relational information.

Protocol. A tuning stage draws configurations at a held-out seed; a confirmation stage re-runs the selection on 15 seeds shared across all arms, so every comparison is paired. We report an exact two-sided sign test alongside the paired t-test and call a result significant only when both agree, with Bonferroni across arms. Every arm receives the same trial budget. Full details, including a cache/RNG confound worth ≈ 5 MCC that dictates run order, are in Appendix A.

3 IS THE MOTIVATION SOUND? δ-HYPERBOLICITY IS AN ARTIFACT

Hyperbolic layers are usually motivated by measuring Gromov δ-hyperbolicity on activations and finding it small, which is read as "the representation is tree-like, so it belongs in hyperbolic space". Before testing the layers we checked that diagnostic, because it is what would tell us where to expect a gain.

It does not survive contact with real activations. Computed on the Euclidean metric, ModernBERT at layer 16 gives $\delta_{\text{eucl}} = 0.0021$, which would indicate an almost perfectly tree-like space. But the same layer has a ratio of maximum to median token norm of 156, and δ is a scale-sensitive statistic: a handful of very-large-norm "attention sink" tokens dominate the Gromov products and drive the estimate toward zero. Recomputing on the *angular* metric $\arccos(\cos(\cdot,\cdot))$, which is invariant to those outliers, gives 0.2127 — against BERT's 0.2114. The two backbones are statistically indistinguishable in tree-likeness, while the Euclidean estimate differs between them by two orders of magnitude. Flan-T5 shows the same pattern with $357\times$ norm outliers.

The practical consequence is that measured δ cannot be used to predict which backbone or which component will benefit from curvature, and any work selecting targets that way is selecting on an

Table 1: One method, four estimators, on CoLA. Each row is the *same* pooler; only the way the score is computed changes. A dash means the knob does not apply — a single-seed row has nothing to average over, and a fixed-config row has nothing to select over. Reading down, the published value sits between our mean and our maximum, and the row that most closely matches how it was computed (single seed, best of a configuration search) *exceeds* it. Only the first row is from Mantri et al. (2026).

Seeds	Aggregated	Selected over	CoLA MCC
Published (Mantri et al., 2026)			
1 (seed 42)	—	best τ of 3	47.49
Ours, same method			
15	mean	—	45.37
15	max	—	48.17
1 (seed 42)	—	best of 40 configs	**50.16**

Table 2: Matched-estimator comparison. Both columns are the best score at GLOT's own seed (42) over GLOT's own grid, 78 configurations per task, so the two are computed the same way. The STS-B residual is unexplained; we test and rule out the leading hypothesis in Appendix F.

Task	ours	published	Δ
CoLA (MCC)	**50.52**	47.49	+3.03
RTE (acc.)	**59.57**	59.21	+0.36
STS-B (Spearman)	82.99	83.86	−0.87

outlier statistic. We therefore do not use it to choose where to apply Stages A–C; we apply all three everywhere and let the paired comparisons decide. We recommend reporting angular δ alongside Euclidean δ, together with the norm-outlier ratio, in any work motivated by measured hyperbolicity.

4 GLOT IS BETTER THAN REPORTED

Mantri et al. (2026) state that "for all experiments, we used a fixed random seed of 42", and their Table 8 reports the best τ per task on the development split they also report. Every published quality number is therefore a single draw, maximised over a search. We report means over 15 seeds. These are not the same estimator, and Table 1 shows that the choice of estimator alone moves CoLA by nearly 5 MCC — more than enough to account for the apparent gap.

Matching their estimator directly — their seed, their grid, 78 configurations per task — reproduces and exceeds two of three tasks (Table 2).

Two consequences follow. GLOT performs better than its own paper claims on CoLA and RTE. And the reported τ-sensitivity curve cannot be read as a τ effect: it spans 7.87 MCC on CoLA, while we measure a spread of 5.46 across seeds at *fixed* τ, and both published rows are flat with a single spike that is the value promoted to their Table 1. Because the estimators differ, every comparison below is against our own baseline under identical conditions, never against the published number.

5 PORTING GLOT TO A NEW BACKBONE

Before comparing geometries we had to establish that each backbone receives a well-formed graph. Eq. 1 compares a cosine to an absolute τ, which is meaningful only if token-cosine scale is comparable across models. It is not, and Table 3 quantifies by how much: the same grid GLOT publishes induces very different graphs on five backbones. This section reports that measurement, and then the more useful result — that GLOT tolerates the variation, while the obvious corrections do not.

RoBERTa is the extreme case: its *tenth*-percentile token cosine is 0.693, above the largest τ in the grid, so no searched configuration makes its graph sparse and GLOT on RoBERTa operates on an

Table 3: Edge density induced by the published grid $\tau \in \{0.1, 0.3, 0.6\}$, with the tenth-percentile token cosine that produces it. Measured directly on 192 CoLA sentences at the **final** layer, over off-diagonal ordered token pairs with padding masked and self-loops excluded; this is a property of the backbone and τ alone, independent of training. Density near 1 means the GNN receives a near-complete graph, and therefore no selective structure.

Backbone	cos p10	τ=0.1	τ=0.3	τ=0.6
TinyLlama-1.1B	0.016	0.762	0.533	**0.073**
ModernBERT-base	0.076	0.866	0.759	0.599
BERT-base	0.098	0.876	0.505	**0.103**
SmolLM2-360M	0.322	0.956	0.908	0.741
RoBERTa-base	**0.693**	1.000	1.000	**0.991**

Table 4: RoBERTa-base, $n = 15$ paired seeds, tuned per arm and per task on RoBERTa (40 trials each, 40 distinct configurations each). `paper_tau` pins the published grid and yields the near-complete graph of Table 3; `density_fix` is the quantile correction. The correction is the worst arm on both tasks, and on CoLA its deficit survives Bonferroni. Notably `density_fix` *won* the tuning stage on CoLA (56.77, best of all five arms) before finishing last over 15 seeds — so it was favoured, not handicapped, by selection.

	CoLA (MCC)			STS-B (Spearman)		
Arm	mean	Δ_{base}	p/n	mean	Δ_{base}	p/n
baseline	**54.60**	*ref*	—	83.89	*ref*	—
`paper_tau`	54.46	-0.14	8/7	**84.21**	$\mathbf{+0.32}$	12/3
A	54.37	-0.23	5/10	83.80	-0.09	5/10
`no_graph`	54.32	-0.29	5/10	83.84	-0.05	7/8
`density_fix`	53.19	$\mathbf{-1.41}$	2/13	83.73	-0.15	5/10

essentially complete graph. SmolLM2 is borderline (0.741 at the sparsest τ). BERT and TinyLlama sit in the regime the threshold was designed for.

The failure is layer-dependent. ModernBERT reaches 0.599 at the final layer but 0.996 at layer 12, which is the layer our fine-tuning experiments use. Mantri et al. (2026) do not state which layer they pool, and Table 3 shows the answer changes the diagnosis — so "which backbones are affected" cannot be settled from the published description alone. We therefore treat RoBERTa as the one unambiguous case.

GLOT tolerates this; density-matching does not. The obvious repair is to replace the absolute threshold with a quantile criterion, retaining the top q fraction of pairs so that realised density is q by construction. We tested it on RoBERTa under the same equal-budget protocol as §6 — 40 tuning trials per arm on RoBERTa itself, per task, then 15 paired confirmation seeds disjoint from the tuning seed. The result is a useful negative for anyone porting this method (Table 4): GLOT at its published threshold is unharmed by the dense graph, while the correction costs accuracy.

The near-complete graph that Table 3 identifies as degenerate is the *best* arm on STS-B (+0.32, 12/3, sign-test $p = 0.035$) and is statistically tied for best on CoLA. Forcing it to be sparse costs 1.41 MCC ($t = -3.44$, 2/13, surviving Bonferroni at $\alpha = 0.0125$). Against `paper_tau` directly the correction loses on both tasks (-1.27 CoLA, -0.48 STS-B, the latter at 2/13 with $p = 0.0074$).

Separating density from scale. We had attributed a $21.71 \rightarrow 27.15$ MCC recovery on ModernBERT/CoLA to a "two-part correction" of density-matching plus median-based norm rescaling, measured at a single seed with the two factors varied together. Table 5 reports the full 2×3 factorial over 5 seeds and four backbone/layer combinations, which separates them. Three things follow.

First, **the choice of rescaling statistic matters far more than whether you rescale**, and this is the only effect in the factorial that is significant on both tests. On ModernBERT layer 12, whose

Table 5: Density × scale factorial, CoLA MCC, mean ± sd over 5 seeds, GLOT's published recipe otherwise unchanged. Columns vary the threshold (absolute, as published, versus density-matched); rows vary token-norm rescaling. *Median* rescaling helps only the backbone/layer with extreme norm outliers; density-matching has no consistent sign; BERT is inert throughout. Note also the variance: ModernBERT layer 12 has a seed sd of 10.16 in the published cell, which is why its single-seed numbers should not be read closely.

Backbone (layer)	Rescaling	$\tau = 0.6$ (pub.)	$q = 0.05$
BERT-base (final) *control*	none	45.25 ±0.97	45.00 ±1.51
	`rms`	45.01 ±1.25	45.03 ±1.18
	`median`	44.98 ±1.00	45.08 ±0.83
ModernBERT-base (L12)	none	18.64 ±10.16	23.22 ±6.17
	`rms`	7.61 ±7.71	19.18 ±2.98
	`median`	24.60 ±7.12	**28.54** ±2.41
ModernBERT-base (final)	none	37.53 ±1.09	35.62 ±6.00
	`rms`	38.64 ±1.48	37.61 ±1.72
	`median`	**38.87** ±1.18	37.57 ±1.79
RoBERTa-base (final)	none	52.39 ±1.98	52.89 ±0.97
	`rms`	52.45 ±1.43	–
	`median`	**52.54** ±1.38	–

mean/median token-norm ratio is 7.66, *mean*-based (`rms`) rescaling costs 11.03 MCC against no rescaling ($t = -5.47$, $0/5$ seeds positive) — the mean is precisely the statistic the outliers corrupt. Density-matching partially rescues that damage (+11.57 within the `rms` condition, $5/0$), which is the only context in which density-matching helps anywhere in this paper.

Second, **median rescaling is directionally positive but not yet established**. It gains 5.96 MCC at absolute τ and 5.32 at $q = 0.05$ on layer 12, and 1.34 at the final layer with $5/0$ seeds positive — but with per-seed standard deviations of 7–10 on layer 12, none of these reaches significance at $n = 5$ ($t = 1.38$, 1.93 and 2.24). We flag a structural limitation of this table: at $n = 5$ the exact two-sided sign test cannot return below $p = 0.0625$, so no contrast here can satisfy the criterion we impose everywhere else. We are extending it to the 15 seeds used in the rest of the paper and report the effect as a lead, not a result.

Third, **density-matching does not transfer**. Its main effect is +4.58 on ModernBERT layer 12 but −1.91 at ModernBERT's final layer, −0.25 on BERT and +0.45 on RoBERTa under this fixed recipe; none is significant, and under the tuned protocol of Table 4 it is significantly harmful on RoBERTa. We therefore withdraw it as a recommendation.

Fourth, the BERT control behaves as a control should: every contrast is at most 0.27 MCC and none is significant, against a seed standard deviation of ≈ 1.0. Our earlier single-seed reading of this control ($45.54 \rightarrow 43.41$) was noise.

6 Does hyperbolic geometry help?

This is the paper's main question. Earlier campaigns of ours searched only graph-construction knobs, inheriting the learning rate, depth, width and projection width from values chosen for a *Euclidean* pooler — an asymmetry that favours the baseline and that produced a false positive we retract in Appendix C. Structured and non-Euclidean layers are known to underperform badly when trained at learning rates tuned for a dense Euclidean layer, so this confound points against the hyperbolic arms and had to be removed before any conclusion was drawn. We therefore repeat the campaign with GLOT's full grid applied identically to every arm: 40 trials per arm, 15 confirmation seeds, 990 runs. Notably *every* arm including the baseline selected $\mathrm{lr} = 2 \times 10^{-4}$; what moved was depth and width.

Table 6 summarises the answer across every setting we ran; Table 7 gives the underlying BERT numbers.

Table 6: Per-stage verdict. Entries are the paired mean difference against GLOT's Euclidean baseline on shared seeds; † marks results significant on both the paired t-test and the exact sign test, ‡ those additionally surviving Bonferroni. "—" was not run. The pattern is consistent: the readout (B) is reliably harmful wherever it appears, message passing (C) is the only component that ever helps, and the graph (A) is flat.

Arm	CoLA (MCC)	STS-B (Spear.)	MRPC (F1)	TinyLlama (Spear.)	Verdict
A	$+0.70$	$+0.03$	$+0.04$	$+0.04$	flat everywhere
B	-1.26	$-1.07^{‡}$	-0.33	—	harmful
C	$+\mathbf{1.42}^{†}$	$+0.20$	-0.32	—	helps on CoLA only
AB	-1.24	$-0.75^{‡}$	-0.28	$-3.56^{‡}$	harmful
AC	$+0.45$	$+\mathbf{0.24}$	-0.15	$+0.21$	flat to slightly positive
BC	$+0.22$	$-1.24^{‡}$	-0.31	$-1.17^{‡}$	harmful
ABC	-0.53	$-0.84^{‡}$	-0.26	$-4.51^{‡}$	harmful

Every arm containing B is negative on STS-B and on the decoder; no arm containing B is positive anywhere. Every arm containing C but not B is positive on CoLA and STS-B.

Table 7: BERT-base, wide search, $n = 15$ paired seeds. Δ_{base} is against GLOT's cosine baseline, Δ_{ng} against the `no_graph` control; "p/n" counts seeds positive/negative against the baseline. Full statistics in Appendix B.

	CoLA (MCC)			STS-B (Spearman)		
Arm	Δ_{base}	p/n	Δ_{ng}	Δ_{base}	p/n	Δ_{ng}
C	$+\mathbf{1.42}$	13/2	$+0.58$	$+0.20$	12/3	$+0.16$
AC	$+0.45$	10/5	-0.39	$+\mathbf{0.24}$	11/4	$+0.20$
`no_graph`	$+0.83$	11/4	*ref*	$+0.04$	7/8	*ref*
A	$+0.70$	11/4	-0.13	$+0.03$	9/5	-0.01
BC	$+0.22$	7/8	-0.61	-1.24	0/15	-1.28
ABC	-0.53	6/9	-1.36	-0.84	1/14	-0.87
AB	-1.24	4/11	-2.07	-0.75	1/14	-0.78
B	-1.26	5/10	-2.10	-1.07	0/15	-1.10

Stage C is the strongest extension. Hyperbolic message passing alone reaches $+1.42$ MCC on CoLA with $13/15$ seeds positive ($t = 4.11$, sign $p = 0.0074$) — significant on both tests, though above the Bonferroni threshold of 0.0063 we apply across eight arms. On STS-B, A+C is the best arm of the nine ($+0.24$, $11/4$), with C second ($+0.20$, $12/3$). Every arm containing the hyperbolic GNN *without* the readout is positive on both tasks.

The hyperbolic readout is harmful, and that is worth reporting separately. Every arm containing Stage B loses 0.75–1.24 Spearman on STS-B with 0–1 of 15 seeds positive; all four survive Bonferroni. The decoder backbone agrees (Appendix D), where the standard deviation also grows monotonically with the number of hyperbolic components ($0.48 \to 3.58$), suggesting the gyromidpoint acts as a variance amplifier rather than a useful inductive bias. Because B and C have opposite signs and similar magnitudes, a study reporting "hyperbolic pooling" as one factor would have averaged them to approximately zero and concluded that geometry does not matter here. It does; the components simply differ.

Scope: Stage C's gain is not yet separable from a no-graph control. We hold our own result to the same standard we apply elsewhere. Alongside the eight arms we ran a `no_graph` control, which also beats the cosine baseline on CoLA ($+0.83$), so part of any arm's advantage may be independent of the graph. Referenced to that control, Stage C's CoLA gain falls to $+0.58$ with 8 seeds positive and 7 negative — an exact sign-test p of 1.00. **We therefore report Stage C as the most promising direction of the three, not as an established improvement**, and we would want a density-matched

Euclidean control at $n \geq 30$ before claiming more. The two arms that *do* separate from the control separate downwards (B and AB, -2.10 and -2.07 MCC, $3/12$, $p = 0.035$), and both are readout arms.

This scope condition applies to GLOT's own graph as well, and there it reads as a strength rather than a limitation: from an empty graph (`no_graph`, realised density 0.000) to a complete one (`paper_tau` on RoBERTa, 0.991), accuracy is unchanged. The same insensitivity holds for the message-passing architecture — swapping GAT for GCN or GIN moves CoLA by at most 0.52 MCC and STS-B by 0.23 (Appendix F). A method this robust to its own hyper-parameters is unusually easy to deploy on a new backbone. GLOT does outperform mean, max and `[CLS]` pooling — by 14–28 points on the authors' own noise diagnostic, in every seed — but a GLOT with *zero* message-passing layers matches the full model there at every noise level, and so does a trainable attention pooler with no graph at all (Appendix E). Taken together, our reading is that much of the benefit is carried by the trained readout, with the graph acting as a token-selection mechanism whose exact construction is not critical. We regard that as a useful localisation for future work on this family, including our own: it says the readout is the component most likely to repay further design effort, which is consistent with Stage C helping and Stage B hurting. Appendix D shows the same ordering on a decoder backbone.

A third task. MRPC, run to the same standard (wide search, $n = 15$ paired seeds, minimum detectable effect 0.317 F1), reproduces the ordering without adding a positive: no arm beats the baseline, `no_graph` is indistinguishable from it ($+0.026$, $9/6$), Stage A is likewise flat ($+0.040$, $8/7$), and every hyperbolic arm is negative (-0.151 to -0.333), none significantly. Stage C's CoLA advantage therefore does not extend to MRPC, which we note as a limit on its generality.

Scope. These results are BERT on CoLA, STS-B and MRPC; RTE is still in progress. Two implementation details in the released code affect anyone reproducing this grid, and we document them in Appendix H so that others do not lose time to them.

7 CONCLUSION

Our question was whether hyperbolic geometry helps graph-based sentence pooling. The answer is that it depends entirely on *which component* receives it, and that the components disagree strongly enough to cancel.

Möbius message passing (C) is the one component that ever helps: $+1.42$ MCC on CoLA with $13/15$ seeds positive, and A+C is the best arm on STS-B. The Einstein gyro-midpoint readout (B) is reliably harmful — negative in every setting we ran, 0–1 of 15 seeds positive on STS-B, surviving Bonferroni, and costing 3.6 Spearman on a decoder backbone. The Poincaré graph (A) is flat everywhere. Averaged as a single "hyperbolic" factor these would report approximately zero, which is what a study that did not decompose the architecture would have concluded.

We are deliberately cautious about C. Its gain does not extend to MRPC, and on CoLA it is not separable from a control that deletes the graph entirely ($8/7$ seeds, sign $p = 1.00$). We therefore report it as the one direction worth pursuing rather than as an established improvement, and we state what would settle it: a density-matched Euclidean control at $n \geq 30$, and Stage C on the decoder backbone, neither of which we have run.

Two lessons generalise beyond this architecture. First, the standard motivation for going hyperbolic — low measured δ-hyperbolicity — is an artifact of norm outliers, and on an outlier-invariant metric the backbones it is used to distinguish are indistinguishable. Second, non-Euclidean layers must be given the same optimiser and architecture budget as the Euclidean baseline before they are judged: our own earlier campaign, which inherited Euclidean hyper-parameters, produced a result we later had to retract. Both points cost us real experiments, and we report them so that the next study of this kind does not repeat them.

REPRODUCIBILITY STATEMENT

All confirmation-stage results use the fixed seed set $\{1, \ldots, 15\}$, shared across every arm so that comparisons are paired; tuning uses a single held-out seed disjoint from that set. Every run appends one row to a per-campaign CSV recording the full configuration, the realised edge density, the per-epoch metrics and the wall-clock time. These logs are released unmodified, including runs that crashed or produced degenerate graphs, and the analysis scripts that produce every table in this paper read those CSVs directly and are released alongside them. Hidden-state caches are built once, before any arm runs; all reported runs then execute against a warm cache. That ordering is load-bearing rather than an optimisation — see Appendix H — and any reproduction that skips it will shift results by several points on CoLA.

REFERENCES

Krishna Sri Ipsit Mantri, Carola-Bibiane Schönlieb, Zorah Lähner, and Moshe Eliasof. Towards improved sentence representations using token graphs. In *International Conference on Learning Representations (ICLR)*, 2026. Verified against extracted PDF text; add URL/pages before submission.

A EXPERIMENTAL PROTOCOL

Frozen backbone, 2 epochs, Adam, learning rate 2×10^{-4}, weight decay 0, batch size 32; the pooler is the only trained component. Confirmation uses the fixed seed set $\{1, \dots, 15\}$ shared across every arm; tuning uses a single disjoint held-out seed.

Matched budget. Every arm receives 40 trials in the wide campaign (10 in the graph-only campaigns). The cosine baseline has a smaller native search space, so we expand its grid to keep selection bias equal. Equal trials is not equal *coverage*: 40 trials cover 6.2% of `no_graph`'s 648-point space but under 0.01% of ABC's 4.98×10^7-point space. This asymmetry disfavours the complex arms, and we note it as a caveat against our own negative results rather than in support of them.

Testing. We report the exact two-sided sign test alongside the paired t-test because evaluation-set quantisation makes the parametric standard error misleadingly small (MRPC accuracy moves in steps of 0.245 on 408 examples). A result counts as significant only when both tests agree. Bonferroni over the 8 arms tested against a common baseline gives $\alpha = 0.0063$.

Cache/RNG confound. The released code precomputes and caches backbone hidden states, and constructs the shuffled `DataLoader` *before* initialising the classifier. On a cache miss the loader is iterated and consumes `torch.randperm` from the global RNG; on a hit it is not. The same seed therefore yields different classifier initialisation depending on cache state — measured at 40.37 (cold) versus 45.54 (warm) MCC on CoLA, about $6\times$ the seed standard deviation. All caches are built before any arm runs. Any reproduction that skips this will shift by several points on CoLA.

B FULL ARM STATISTICS

Table 7 reports point estimates and seed counts. The corresponding paired statistics on STS-B are: B $\bar{d} = -1.065$, $t = -17.00$, sign $p = 6 \times 10^{-5}$; BC -1.242, $t = -8.30$, $p = 6 \times 10^{-5}$; ABC -0.837, $t = -6.36$, $p = 0.00098$; AB -0.747, $t = -5.66$, $p = 6 \times 10^{-5}$. All four survive Bonferroni. On CoLA, C gives $\bar{d} = +1.416$, $t = 4.11$, sign $p = 0.0074$ — significant on both tests but above $\alpha = 0.0063$. The minimum detectable effect at $n = 15$ is 0.225 (STS-B) and 1.153 (CoLA).

C A RETRACTED RESULT OF OUR OWN, AND WHY

Under the graph-only search, Stage A on STS-B gave $\bar{d} = +0.223$, $t(14) = 6.20$, sign $p = 6 \times 10^{-5}$, $15/15$ seeds positive, and replicated on 12 held-out seeds. It does not survive the wide search: the same comparison gives $\bar{d} = +0.028$, $t = 0.69$, $9/5$ seeds. The effect was real but its cause was not geometry — the baseline had been pinned at $K = 2$, $h = 128$ while nothing else was, and it recovers the entire gap once it can choose $K = 4$. We report this because it is the clearest illustration of the paper's own point: a pooling comparison is only as trustworthy as the budget given to its control.

D DECODER BACKBONE

TinyLlama-1.1B on STS-B, $n = 15$ shared seeds. This backbone was selected over SmolLM2 on measured geometry: Table 3 puts it in BERT's regime (0.073 against BERT's 0.103 at $\tau = 0.6$) whereas SmolLM2 sits at 0.741. The minimum detectable effect is 0.503.

Densities below are recovered from training telemetry, which uses the defective statistic described in Appendix H: it reports $(|E_{\text{off}}| + n)/(n(n-1))$, so an empty graph reads $1/(n-1)$ rather than 0. We subtract that offset, which the `no_graph` arm identifies exactly ($0.095 \to 0$) and which is internally consistent across all seven arms. These are STS-B sentences and so are not directly comparable with the CoLA measurements of Table 3.

Arm	mean	std	density	Δ	sign p	pos/neg
AC	80.15	0.54	0.027	$+0.205$	0.302	10/5
A	79.99	0.51	0.099	$+0.043$	0.607	9/6
`no_graph`	79.95	0.48	0.000	$+0.001$	1.000	8/7
baseline	79.95	0.49	0.026	*ref*	—	—
BC	78.78	0.95	0.101	-1.165	0.00098	1/14
AB	76.39	1.87	0.100	-3.556	6×10^{-5}	0/15
ABC	75.44	3.58	0.100	-4.506	6×10^{-5}	0/15

The ordering matches BERT: `no_graph` is indistinguishable from the baseline ($\Delta = +0.001$, $p = 1.00$) despite the baseline having a well-conditioned graph, and the readout arms are significantly harmful. Standard deviation grows monotonically with the number of hyperbolic components ($0.48 \to 0.95 \to 1.87 \to 3.58$), consistent with these operators acting as variance amplifiers rather than as useful inductive bias. Realised density spans 0.000–0.101 across arms, so a density-matched Euclidean control is still needed to fully separate sparsity from geometry; this matters least for the two arms that matter most, since AC sits at 0.122 against the baseline's 0.121.

E DIAGNOSTIC STRESS TEST

We reproduce the signal-in-noise diagnostic of Mantri et al. (2026): a short signal phrase carrying a logical dependency, injected into a sequence of random distractor words, with the distractor ratio swept from 20% to 90%. BERT backbone, mean $\pm$ std over 5 seeds; the original reports a single seed.

Arm	20%	50%	80%	90%
`no_graph`	**98.0** $\pm$1.3	**94.8** $\pm$5.7	**92.5** $\pm$5.0	84.4 $\pm$10.1
AC	95.0 $\pm$3.4	88.2 $\pm$11.2	90.8 $\pm$5.8	**88.5** $\pm$8.9
C	95.0 $\pm$3.3	87.6 $\pm$11.2	90.8 $\pm$5.9	88.4 $\pm$8.8
A	97.0 $\pm$1.3	91.9 $\pm$4.1	91.0 $\pm$2.7	84.9 $\pm$9.1
baseline	96.0 $\pm$0.5	90.2 $\pm$5.0	88.1 $\pm$4.7	79.8 $\pm$10.8
B	95.4 $\pm$1.4	88.7 $\pm$2.8	83.8 $\pm$13.8	75.5 $\pm$14.9
Mantri et al. (2026) Table 7, BERT row (single seed):				
GLOT	97.2	97.0	97.8	98.8

Two observations. The ordering again places `no_graph` at or above the baseline at every ratio, in the experiment designed specifically to demonstrate the graph's value.

Second, we do not reproduce the published GLOT numbers: at 90% distractors the original reports 98.8 where we measure 79.8 ± 10.8 for the same method, and their curve is flat or rising under increasing noise where ours degrades monotonically. We flag this as unresolved rather than as a refutation — the generation procedure is underspecified in the original (vocabulary source and signal-phrase templates are not given), so the gap is plausibly task construction rather than method.

Competing poolers. Mantri et al. (2026)'s claim is comparative: GLOT holds up under noise where other poolers collapse. We ran those poolers on our own generator, 5 shared seeds each, together with a GLOT variant at $K = 0$ that keeps the adaptive scorer but performs no message passing — the parameter-matched control for whether the benefit is relational or simply capacity. Deltas below are paired against full GLOT on the same seeds, which is a comparison internal to our data and so does not depend on matching the published values.

Pooler	type	20%	50%	80%	90%
GLOT (K=0)	trained	95.7	91.0	89.3	88.1
AdaPool	trained	95.1	90.8	86.5	83.6
GLOT	trained	96.0	90.2	88.2	79.7
Mean	static	73.5	63.2	64.0	57.4
`[CLS]`	static	70.0	61.4	62.1	65.4
Max	static	63.0	55.2	53.1	51.8
paired Δ vs GLOT *(pos/neg of 5 seeds, exact sign test):*					
K=0		-0.3 (2/3, 1.00)	$+0.8$ (2/3, 1.00)	$+1.0$ (4/1, 0.38)	$+8.4$ (3/2, 1.00)
AdaPool		-0.8 (1/4, 0.38)	$+0.7$ (3/1, 0.63)	-1.8 (2/3, 1.00)	$+3.9$ (4/1, 0.38)
static		-14 to -35, $0/5$ at every ratio ($p = 0.062$, the floor at $n = 5$)			

Two conclusions, and one caution about what this table cannot support.

The trained/static split is unambiguous and holds in every seed: mean, `[CLS]` and max lose 14–35 points to GLOT at every ratio, $0/5$ seeds positive. GLOT's advantage over set-based pooling is real.

Within the trained group, however, nothing separates. $K = 0$ — no message passing, no graph consumed at all — is statistically indistinguishable from full GLOT at every ratio, and so is AdaPool, which has no token graph whatsoever. We emphasise that we do *not* claim $K = 0$ is better: its $+8.4$ at 90% rests on $3/5$ seeds with a sign-test p of 1.00 and per-seed sd of 5.8 and 9.7. The claim is the null one, and it is the relevant one: the diagnostic designed to demonstrate that relational reasoning over the token graph confers noise robustness does not distinguish a model that performs that reasoning from two models that do not.

Caution: our trained poolers do not match theirs. The divergence between our numbers and the published ones is not uniform, and the pattern matters. The static poolers agree closely (-2.2 to $+5.6$ across all twelve cells). Both *trained* poolers diverge instead, in opposite directions, and the divergence grows with noise: at 90% our AdaPool is 22.0 points *above* the published value while our GLOT is 19.1 points *below* it. Consequently the 37-point margin their Table 7 reports for GLOT over AdaPool at 90% becomes -3.9 in our hands. We therefore cannot claim to reproduce their setup for trained poolers, and we do not use the published values as a baseline anywhere; only the paired within-data deltas above carry weight. The comparison their claim rests on is exactly the one that fails to replicate, and their Appendix B.4 does not specify the vocabulary source, the signal-phrase templates, or the pooler training budget, so we cannot localise the cause.

F GNN BACKBONE AND TRAINING RECIPE

GNN backbone. Every arm in the main campaigns pins `gnn_type=gat`, so a reviewer may reasonably ask whether "the graph carries no information" is really "GAT extracts none". It is not: over 5 seeds with everything else at the published recipe, CoLA gives GAT 45.38 ± 1.19, GCN 45.14 ± 1.78, GIN 44.86 ± 1.33, and STS-B gives 82.69 ± 0.34, 82.46 ± 0.27, 82.47 ± 0.32. The spread is 0.52 MCC and 0.23 Spearman, well inside the seed noise, and GAT is the only one of the three that consumes edge attributes at all. Message-passing architecture is not where the variation lives either.

The STS-B residual is not a recipe mismatch. §4 leaves an unexplained -0.87 on STS-B. The released README trains differently from the paper's Appendix B.2 (3 epochs, `scorer_hidden` 256, $\tau = 0.8$ versus 2, 128, 0.6), so we ran the $2 \times 2 \times 2$ decomposition at the authors' seed 42. Epoch count is the only factor that moves anything ($82.36 \rightarrow 82.66$ for $2 \rightarrow 3$ epochs); τ and `scorer_hidden` are inert to within 0.04. Over 5 seeds the two full recipes are indistinguishable (82.69 ± 0.34 against 82.69 ± 0.23). The best cell we obtain, 82.66, still sits 1.20 below the published 83.86, so the residual is not explained by the recipe discrepancy and we leave it open.

G MEASUREMENT ARTIFACTS

Tuning maxima are inflated. With T trials and per-seed noise σ, the expected maximum is inflated even for an arm identical to the baseline: ≈ 1.83 here (STS-B/BERT, $\sigma = 1.19$) and ≈ 0.83 (STS-B/TinyLlama, $\sigma = 0.54$). Several apparent leads in this project were smaller than this and did not survive confirmation. Relatedly, the noise floor must be measured for the configuration being confirmed: we initially assumed CoLA's $\sigma = 0.81$ from a layer-12 probe, where the true value at layer 8 is 2.09.

The RoBERTa campaign gives the sharpest illustration, because the inflation there is large enough to *reverse the ranking*. Mean drop from best-tuning score to the 15-seed confirmation mean is 2.08 MCC on CoLA and 0.48 Spearman on STS-B, and no arm's rank is preserved: `density_fix` is first of five at the tuning maximum (56.77) and last of five over 15 seeds (53.19). Any protocol that reports the tuning maximum would have concluded the opposite of what we report in §5.

Layer selection overfits. Selecting a backbone layer on one task does not transfer: layer 8 beat layer 12 by $+4.62$ MCC on CoLA but by only $+0.25$ on STS-B, and lost on RTE (-0.36) and MRPC (-0.46).

H NOTES FOR REPRODUCERS

Three details of the released implementation cost us time and are recorded here so they do not cost anyone else's. None affects the conclusions of the original paper.

- `jk_mode=lstm` is accepted by the argument parser but raises at runtime, and the advertised options `mean` and `none` are not implemented. The effective space is {`cat`, `max`}.
- Thresholding Poincaré *distance* at an absolute radius yields an empty graph on real features. BERT token norms (≈ 14.7) saturate $\exp_0^c$ to radius 1.0 exactly in `float32`, and real pairwise geodesic distances land in $[8.85, 11.76]$, so any absolute threshold below ≈ 9 gives zero edges. The diagnostic signature is a score that is bit-identical across an entire threshold grid. We use a quantile threshold throughout for this reason.
- The logged edge-density statistic counts self-loops in its numerator but excludes the diagonal from its denominator, so it reports $|E|/(n(n-1))$ with $|E|$ including the n self-loops. A complete graph therefore logs $n/(n-1) > 1$ and an *empty* graph logs $1/(n-1) \approx 0.10$ at CoLA's typical length rather than 0. This is a reporting defect only — no training path consumes the statistic, so no score is affected — but it is a trap for exactly the audit this paper performs. Two consequences are worth recording. Our `no_graph` control logs density 0.108 on BERT/CoLA, which looks like a live graph and is in fact the empty-graph signature $1/(n-1)$; and the corrected BERT baseline density, $0.2074 - 0.10 \approx 0.104$, independently reproduces the 0.103 measured directly in Table 3, which is what confirms the correction. Any density we quote is re-measured with the standalone instrument described in Table 3, never taken from training telemetry.

I STRUCTURAL (NON-GEOMETRIC) GRAPH VARIANTS

Since CoLA, MRPC and RTE are flat for every *geometric* arm, we also asked whether a *structural* prior helps where a similarity prior does not: positional edges connecting tokens with $|i - j| \leq w$, positional-only graphs, their union with Stage A, and k-NN graphs.

Arm	trials	best	mean
baseline	10	45.32	43.42
POS	10	47.08	44.50
POS_ONLY	4	46.80	44.38
A_POS	10	44.87	43.48
KNN	4	45.32	44.84

These are tuning-stage maxima and are not results. At CoLA's confirmation-stage noise level ($\sigma = 2.09$) the expected inflation of a best-of-10 maximum is ≈ 3.22 MCC, which exceeds every gap in the table — including POS's apparent $+1.76$ lead. We report the numbers for completeness and because the arms are implemented and released, but we draw no conclusion from them. Confirming the structural arms on 15 paired seeds is the most obvious extension of this work that we did not run.

J LIMITATIONS

- **Stage C is a direction, not a result.** Its CoLA gain does not survive referencing against the `no_graph` control (8/7 seeds, sign $p = 1.00$), does not extend to MRPC, and was never run on the decoder backbone. A density-matched Euclidean control at $n \geq 30$ would settle it; we did not run one.
- **Our own numbers select the epoch on the split we report**, because GLUE test labels are not public. This is a smaller-scale version of the practice we characterise in §4: we select over epochs and configurations, the original additionally selects over τ at one seed. Splitting the development set into selection and reporting halves would remove the objection entirely, and we regard it as the single most valuable improvement to this protocol.
- **One reproduction residual is unexplained.** At the authors' seed and grid we exceed the published CoLA and RTE numbers, but STS-B remains 0.87 short under every estimator we tried. We tested the leading hypothesis — that the released README trains differently from the paper (3 vs. 2 epochs, hidden 256 vs. 128, $\tau = 0.8$ vs. 0.6) — and ruled it out (Appendix F). The residual stands.
- **The scale factorial is underpowered.** At $n = 5$ the exact sign test cannot fall below $p = 0.0625$, so no contrast in Table 5 can meet the both-tests criterion we impose elsewhere. The median-rescaling effect is directionally positive on both ModernBERT layers but not established.
- **Non-Euclidean layers may need structure-aware optimisation.** Structured and hyperbolic layers are known to underperform when trained at learning rates and initialisations tuned for dense Euclidean layers. We addressed the learning-rate half of this objection by giving every arm the full grid — and every arm, including the baseline, selected $\mathrm{lr} = 2 \times 10^{-4}$ — but we did not implement per-manifold Riemannian optimisation or curvature-aware initialisation. This is the strongest available objection to our negative result on Stage B, and we state it rather than only acknowledging it.
- **Scope.** Results cover BERT, RoBERTa, ModernBERT, SmolLM2 and TinyLlama on CoLA, STS-B, MRPC and a synthetic stress task. RTE was still running at submission. We report no MTEB or IMDB results, which the original paper does.